Solutions

Maths

Q1. The value of the expression $\sin\left(\frac{\pi}{12}\right)\sum_{k=1}^{6}\frac{1}{\sin\left(\frac{k\pi}{12}\right)\sin\left(\frac{(k+1)\pi}{12}\right)}$ is equal to:

(A) 2 (B) $2\sqrt{3}$ (C) 4 (D) $4\sqrt{3}$

Ans. C

Sol. Let $\theta = \frac{\pi}{12}$.

Using the identity

$$\frac{\sin\theta}{\sin(k\theta)\sin((k+1)\theta)} = \cot(k\theta) - \cot((k+1)\theta)$$

The given expression can be written as:

$$S = \sum_{k=1}^{6}(\cot(k\theta) - \cot((k+1)\theta)) = \cot\theta - \cot 7\theta$$

Substituting $\theta = \frac{\pi}{12}$:

$$S = \cot\left(\frac{\pi}{12}\right) - \cot\left(\frac{7\pi}{12}\right)$$

$$\cot\left(\frac{7\pi}{12}\right) = \cot\left(\frac{\pi}{2} + \frac{\pi}{12}\right) = -\tan\left(\frac{\pi}{12}\right)$$

Thus,

$$S = \cot\left(\frac{\pi}{12}\right) + \tan\left(\frac{\pi}{12}\right) = 4$$

Hence, the correct option is (C).

Q2. Consider the trigonometric inequality $\sqrt{3}\tan x - 2\sin x \le 0$ where $x \in [0, 2\pi) \setminus \left\{\frac{\pi}{2}, \frac{3\pi}{2}\right\}$. If the complete solution set of this inequality is $[0, a] \cup (b, c] \cup (d, e]$, where $0 < a < b < c < d < e < 2\pi$, then the value of $a + b + c + d + e$ is equal to :

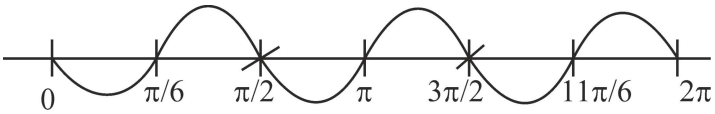
(A) 4π (B) 5π (C) $\frac{9\pi}{2}$ (D) $\frac{11\pi}{2}$

Ans. B

Sol. Rewrite the given inequality as: $\tan x\left(\sqrt{3} - 2\cos x\right) \le 0$

$$\Rightarrow \frac{\sin x\left(\frac{\sqrt{3}}{2} - \cos x\right)}{\cos x} \leqslant 0.$$

Critical points are $x = 0, \frac{\pi}{6}, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \frac{11\pi}{6}$



Thus, the complete solution set is: $\left[0, \frac{\pi}{6}\right] \cup \left(\frac{\pi}{2}, \pi\right] \cup \left(\frac{3\pi}{2}, \frac{11\pi}{6}\right]$

$a = \frac{\pi}{6}, \quad b = \frac{\pi}{2}, \quad c = \pi, \quad d = \frac{3\pi}{2}, \quad e = \frac{11\pi}{6}$

Therefore, $a + b + c + d + e = \frac{\pi}{6} + \frac{\pi}{2} + \pi + \frac{3\pi}{2} + \frac{11\pi}{6} = 5\pi$

Correct Option: (B)

Q3. Let α and β be the roots of the quadratic equation $x^2 - 2x - 1 = 0$. For any positive integer n, let $a_n = \alpha^n - \beta^n$. If the system of homogeneous linear equations

$a_8 x + a_7 y + a_6 z = 0$

$a_7 x + a_6 y + a_5 z = 0$

$\lambda x + 2y + z = 0$

has a non-trivial solution, then the value of λ is:

(A) 3 (B) 5 (C) 1 (D) 4

Ans. B

Sol. Since α and β are the roots of $x^2 - 2x - 1 = 0$, by Newton's Formula for $a_n = \alpha^n - \beta^n$, we have:

$a_n - 2a_{n-1} - a_{n-2} = 0 \implies a_n - 2a_{n-1} - a_{n-2} = 0$

For the given system of homogeneous equations to have a non-trivial solution, the determinant of the coefficient matrix must be zero:

$$D = \begin{vmatrix} a_8 & a_7 & a_6 \\ a_7 & a_6 & a_5 \\ \lambda & 2 & 1 \end{vmatrix} = 0$$

Applying column operation $C_1 \to C_1 - 2C_2 - C_3$:

$$D = \begin{vmatrix} a_8 - 2a_7 - a_6 & a_7 & a_6 \\ a_7 - 2a_6 - a_5 & a_6 & a_5 \\ \lambda - 2(2) - 1 & 2 & 1 \end{vmatrix} = 0$$

Using Newton's relation, $a_8 - 2a_7 - a_6 = 0$ and $a_7 - 2a_6 - a_5 = 0$.

Thus: $D = \begin{vmatrix} 0 & a_7 & a_6 \\ 0 & a_6 & a_5 \\ \lambda - 5 & 2 & 1 \end{vmatrix} = 0$

Expanding along the first column:

$(\lambda - 5)(a_7 a_5 - a_6^2) = 0$

Since $a_7 a_5 - a_6^2 = -(\alpha\beta)^5(\alpha - \beta)^2$

$= -(-1)^5(2^2 - 4(-1)) = 8 \neq 0$, we get:

$\lambda - 5 = 0 \implies \lambda = 5$

Q4. Let a, b, c be three positive real numbers in Geometric Progression. Let A and H denote the Arithmetic Mean and Harmonic Mean of a, b, c respectively. If $A + H = 13$ and $A - H = 5$, then the value of $(a - c)^2$ is equal to:

(A) 297 (B) 315 (C) 417 (D) 585

Ans. A

Sol. Let a, b, c be in Geometric Progression

The middle term is b, so $ac = b^2$.

The Arithmetic Mean A, Geometric Mean G, and Harmonic Mean H of a, b, c are given by:

$A = \dfrac{a+b+c}{3}$

$G = (abc)^{1/3} = (b^3)^{1/3} = b$

$H = \dfrac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \dfrac{3abc}{ab + bc + ca}$

$= \dfrac{3b^3}{b(a+c) + b^2} = \dfrac{3b^2}{a+b+c}$

$A \cdot H = \left(\dfrac{a+b+c}{3}\right) \cdot \left(\dfrac{3b^2}{a+b+c}\right) = b^2 = G^2$

From the given equations: $A + H = 13$, $A - H = 5$ $\Rightarrow$ A = 9, H = 4 and G = b = 6

Since $A = \dfrac{a+b+c}{3} = 9$, we have: $a + b + c = 27 \implies a + c = 27 - 6 = 21$

Also, $ac = b^2 = 36$.

Now, evaluating $(a - c)^2$: $(a - c)^2 = (a + c)^2 - 4ac = 21^2 - 4(36)$

$= 441 - 144 = 297$

Q5. The sum of all real values of $x \in [0, 2\pi]$ satisfying the equation $\sin x + \cos x = \sqrt{1 + 2\cos^2 x}$ is equal to :

(A) $\dfrac{3\pi}{4}$ (B) $\dfrac{7\pi}{2}$ (C) $\dfrac{11\pi}{4}$ (D) $\dfrac{\pi}{4}$

Ans. A

Sol. Given equation:

$\sin x + \cos x = \sqrt{1 + 2\cos^2 x}$

Since the right-hand side is a principal square root, we must have:

$\sin x + \cos x \geq 0$ --- (1)

Squaring both sides we get :

$2\cos x(\sin x - \cos x) = 0$

$\cos x = 0 \implies x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$ in $[0, 2\pi]$.

$\sin x = \cos x \implies \tan x = 1 \implies x = \dfrac{\pi}{4}, \dfrac{5\pi}{4}$ in $[0, 2\pi]$.

Thus, the valid solutions in $[0, 2\pi]$ are $x = \dfrac{\pi}{4}$ and $x = \dfrac{\pi}{2}$. (From 1.)

Sum of solutions = $\frac{\pi}{4}+\frac{\pi}{2}=\frac{3\pi}{4}$.

Q6. Let H be the harmonic mean of the numbers 4 and 12. A sequence $\{a_n\}$ is defined such that $a_1 = H$ and $a_{n+1} = 2a_n + 3$ for all $n \geq 1$. If $b_n = a_n + \lambda$ (where λ is a constant) forms a geometric progression, then the value of b_5 is equal to :

(A) 72 (B) 144 (C) 288 (D) 96

Ans. B

Sol. The harmonic mean H of 4 and 12 is given by:

$H = \frac{2 \times 4 \times 12}{4+12} = \frac{96}{16} = 6$

Thus, $a_1 = 6$.

Given recurrence relation:

$a_{n+1} = 2a_n + 3$

Adding 3 to both sides:

$a_{n+1} + 3 = 2(a_n + 3)$

Comparing this with $b_n = a_n + \lambda$, we get $\lambda = 3$ and $b_{n+1} = 2b_n$.

Hence, $\{b_n\}$ is a geometric progression with common ratio $r = 2$ and first term:

$b_1 = a_1 + 3 = 6 + 3 = 9$

The 5th term b_5 is given by:

$b_5 = b_1 \cdot r^4 = 9 \cdot 2^4 = 9 \cdot 16 = 144$

Q7. Let $P(x) = x^3 + (a-2)x^2 - (a+1)x + 2 = 0$ be a cubic equation, where $a \in \mathbb{R}$. If $x = 1$ is a root of $P(x) = 0$, then the complete set of values of a for which the remaining roots of $P(x) = 0$ contain exactly one root in the open interval $(1, 3)$ is :

(A) $\left(-\frac{4}{3}, 2\right)$ (B) $\left(-2, \frac{4}{3}\right)$ (C) $\left(-\frac{4}{3}, \frac{2}{3}\right)$ (D) $(-\infty, 2)$

Ans. A

Sol. Given the cubic polynomial:

$P(x) = x^3 + (a-2)x^2 - (a+1)x + 2 = 0$

Since $x = 1$ is a root, we factorize $P(x)$ as:

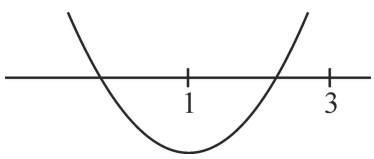
$P(x) = (x-1)(x^2 + (a-1)x - 2) = 0$

The remaining roots are the solutions of the quadratic equation:

$g(x) = x^2 + (a-1)x - 2 = 0$

g(x) = 0 must have a root in (1, 3)

Case-I



g(1).g(3) < 0 $\Rightarrow a \in \left(-\frac{4}{3}, 2\right)$ .....(1)

Case-II : If one roots of g(x) = 0 is either 1 or 3 and other lies in (1, 3)

For $g(x) = 0$, the product of roots is $\frac{-2}{1} = -2 < 0$.

This implies that $g(x) = 0$ has one positive root and one negative root.

Hence its not possible that one root of g(x) = 0 is either 1 or 3 and other lies in (1, 3)

$a \in \phi$ .....(2)

Taking the union of both conditions:

$a \in \left(-\frac{4}{3}, 2\right)$

Q8. Let R be the 2×2 matrix representing a counter-clockwise rotation of the 2D plane about the origin by an angle of $\frac{\pi}{3}$, and let P be the 2×2 matrix representing a reflection about the line $y = x$. If the system of linear equations $(2R - \lambda P)\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ has no solution, then the positive real value of λ is :

(A) 1 (B) 2 (C) $\sqrt{3}$ (D) 4

Ans. B

Sol. The rotation matrix R for an angle $\theta = \frac{\pi}{3}$ is given by:

$$R = \begin{bmatrix} \cos(\pi/3) & -\sin(\pi/3) \\ \sin(\pi/3) & \cos(\pi/3) \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix}$$

The reflection matrix P across the line $y = x$ is :

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Now, compute the coefficient matrix $A = 2R - \lambda P$:

$$A = 2\begin{bmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} - \lambda \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -(\lambda + \sqrt{3}) \\ \sqrt{3} - \lambda & 1 \end{bmatrix}$$

For the system of linear equations $A\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ to have no solution, the determinant of matrix A must be zero:

$\det(A) = 1 \cdot 1 - (-(\lambda + \sqrt{3}))(\sqrt{3} - \lambda) = 0$

$1 + (\lambda + \sqrt{3})(\sqrt{3} - \lambda) = 0$

$1 + (3 - \lambda^2) = 0 \implies \lambda^2 = 4 \implies \lambda = \pm 2$

For $\lambda = 2$, the system becomes:

$$\begin{cases} x - (2 + \sqrt{3})y = 1 \\ (\sqrt{3} - 2)x + y = 2 \end{cases}$$

This system is inconsistent and has no solution.

Thus, the positive value of λ is 2.

Q9. Consider the system of homogeneous linear equations in x, y, z:

$(2\cos\theta)x + y + z = 0$

$x + (2\cos\theta)y + z = 0$

$x + y + (2\cos\theta)z = 0$ where $\theta \in [0, 2\pi)$.

Let S_1 be the set of values of θ for which the set of non-trivial solutions forms a line passing through the origin in $\mathbb{R}^3$. Let S_2 be the set of values of θ for which the set of solutions forms a plane passing through the origin in $\mathbb{R}^3$.

Then the sum of all elements in the set $S_1 \cup S_2$ is equal to:

(A) $\frac{5\pi}{3}$ (B) 2π (C) 3π (D) $\frac{7\pi}{2}$

Ans. C

Sol. For non trivial solution

$$\begin{vmatrix} 2\cos\theta & 1 & 1 \\ 1 & 2\cos\theta & 1 \\ 1 & 1 & 2\cos\theta \end{vmatrix} = 0$$

$(2\cos\theta + 2)(2\cos\theta - 1)^2 = 0$

$= 8(\cos\theta + 1)\left(\cos\theta - \frac{1}{2}\right)^2 = 0$

$\cos\theta = -1 \implies \theta = \pi$ (since $\theta \in [0, 2\pi)$).

$\cos\theta = \frac{1}{2} \implies \theta = \frac{\pi}{3}, \frac{5\pi}{3}$.

Case-I : $\cos\theta = -1$ then system of equations becomes

$(-2)x + y + z = 0$ .....(1)

$x + (-2)y + z = 0$ .....(2)

$x + y + (-2)z = 0$ .....(3)

Plane in eq. 3 is passing through line of intersection of planes in eq. (1) and (2).

Hence solutions lies on a line passing through origin

$S_1 = \{\pi\}$

Case-II : $\cos\theta = 1/2$ then system of equations becomes

$x + y + z = 0$

$x + y + z = 0$

$x + y + z = 0$

Hence solution lies in plane x + y + z = 0

$S_2 = \left\{\frac{\pi}{3}, \frac{5\pi}{3}\right\}$

$S_1 \cup S_2 = \left\{\pi, \frac{\pi}{3}, \frac{5\pi}{3}\right\}$

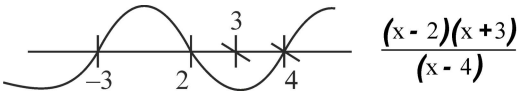
Q10. Let $P(x) = x^3 - 2x^2 - 9x + 18$ be a cubic polynomial. The complete set of real values of x satisfying the inequality $\dfrac{P(x)}{x^2 - 7x + 12} \leq 0$ is :

(A) $(-\infty, -3] \cup [2, 4)$ (B) $(-\infty, -3] \cup [2, 3) \cup (3, 4)$ (C) $(-\infty, -3] \cup (2, 3) \cup (3, 4)$

(D) $[-3, 2] \cup (3, 4)$

Ans. B

Sol. $$\frac{p(x)}{x^2 - 7x + 12} = \frac{(x-2)(x+3)}{(x-4)}, x \neq 3, 4$$



$x \in (-\infty, -3] \cup [2, 3) \cup (3, 4)$.

Q11. Let a and b be two positive real numbers such that $a + b = 10$. The maximum value of the product ab is M. If H is the Harmonic Mean of a and b when ab attains its maximum value, then the value of $M + H$ is equal to:

(A) 20 (B) 25 (C) 35 (D) 30

Ans. D

Sol. Using the AM-GM inequality for positive real numbers a and b:

$\dfrac{a+b}{2} \geq \sqrt{ab}$

Substitute $a + b = 10$:

$\dfrac{10}{2} \geq \sqrt{ab} \implies 5 \geq \sqrt{ab} \implies ab \leq 25$

Thus, the maximum value of ab is $M = 25$.

The equality in AM-GM holds when $a = b$.

Since $a + b = 10$ and $a = b$, we get $a = 5$ and $b = 5$.

Now, the Harmonic Mean H of $a = 5$ and $b = 5$ is given by:

$H = \dfrac{2ab}{a+b} = \dfrac{2(5)(5)}{5+5} = \dfrac{50}{10} = 5$

Therefore, $M + H = 25 + 5 = 30$

Q12. Let $P = \dfrac{1}{3}\begin{bmatrix} 1 & 2 & a \\ 2 & 1 & b \\ 2 & -2 & c \end{bmatrix}$ be an orthogonal matrix such that $\det(P) = 1$. Let $A = P - kI_3$, where $k \in \mathbb{R}$ and I_3 is the identity matrix of order 3. If the system of homogeneous linear equations $AX = 0$ has a non-trivial solution, then a possible real value of k is :

(A) 1 (B) 2 (C) 3 (D) 0

Ans. A

Sol. Since P is an orthogonal matrix with $\det(P) = 1$, its columns C_1, C_2, C_3 are orthonormal such that $C_3 = C_1 \times C_2$.

Given $C_1 = \frac{1}{3}\begin{bmatrix}1\\2\\2\end{bmatrix}$ and $C_2 = \frac{1}{3}\begin{bmatrix}2\\1\\-2\end{bmatrix}$, we compute:

$$C_3 = \frac{1}{9}\begin{bmatrix}1\\2\\2\end{bmatrix} \times \begin{bmatrix}2\\1\\-2\end{bmatrix} = \frac{1}{3}\begin{bmatrix}-2\\2\\-1\end{bmatrix}$$

Thus, $a = -2$, $b = 2$, and $c = -1$.

The system $AX = 0$ has a non-trivial solution if and only if $\det(A) = 0$, which implies:

$\det(P - kI_3) = 0$

Substituting a, b, c into $\det(P - kI_3) = 0$:

$$\begin{vmatrix} \frac{1}{3} - k & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} - k & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} - k \end{vmatrix} = 0$$

Let $x = 3k$, then:

$$\begin{vmatrix} 1 - x & 2 & -2 \\ 2 & 1 - x & 2 \\ 2 & -2 & -1 - x \end{vmatrix} = 0$$

$$\implies (3 - x)\left[(x + 1)^2 + 8\right] = 0$$

The real root is $x = 3 \implies 3k = 3 \implies k = 1$.

Q13. Let $A = \begin{pmatrix}1 & 1 & 2\\0 & 1 & 1\\0 & 0 & 1\end{pmatrix}$ be a 3×3 matrix. If $B = \text{adj}(A^{10})$, then the sum of all elements of the matrix B is equal to:

(A) 17 (B) 18 (C) 20 (D) 35

Ans. B

Sol. Write $A = I + N$, where $I = \begin{pmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{pmatrix}$ and $N = \begin{pmatrix}0 & 1 & 2\\0 & 0 & 1\\0 & 0 & 0\end{pmatrix}$.

Calculating powers of N:

$$N^2 = \begin{pmatrix}0 & 0 & 1\\0 & 0 & 0\\0 & 0 & 0\end{pmatrix}, \quad N^3 = O$$

Since $N^3 = O$, by binomial expansion for commuting matrices:

$$A^{10} = (I+N)^{10} = I + 10N + \frac{10\times 9}{2}N^2$$

$$A^{10} = \begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix} + \begin{pmatrix}0&10&20\\0&0&10\\0&0&0\end{pmatrix} + \begin{pmatrix}0&0&45\\0&0&0\\0&0&0\end{pmatrix}$$

$$= \begin{pmatrix}1&10&65\\0&1&10\\0&0&1\end{pmatrix}$$

Now, let $M = A^{10}$.

$$B = \text{adj}(M) = \begin{pmatrix}1&-10&35\\0&1&-10\\0&0&1\end{pmatrix}$$

Sum of all elements of matrix B:

Sum $= 1 + (-10) + 35 + 0 + 1 + (-10) + 0 + 0 + 1$

$= 18$

Q14. Let x, y, z be distinct real numbers and $k \in \mathbb{R}$. Consider the matrix $M(k)$ defined as:

$$M(k) = \begin{pmatrix} 1+x+x^2 & 1+y+y^2 & 1+z+z^2 \\ 1+3x+9x^2 & 1+3y+9y^2 & 1+3z+9z^2 \\ 1+(k^2-k)x+k^2x^2 & 1+(k^2-k)y+k^2y^2 & 1+(k^2-k)z+k^2z^2 \end{pmatrix}$$

If the third row of $M(k)$ can be expressed as a linear combination of its first two rows, then the sum of all possible real values of k is:

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $-\frac{4}{3}$ (D) 2

Ans. A

Sol. $\because R_3$ is given by $\lambda R_1 + \mu R_2$

$\Rightarrow \lambda + \mu = 1$ .....(1)

$\Rightarrow \lambda + 3\mu = k^2 - k$ .....(2)

$\Rightarrow \lambda + 9\mu = k^2$ .....(3)

$$\begin{vmatrix}1&1&1\\1&3&k^2-k\\1&9&k^2\end{vmatrix} = 0$$

$\Rightarrow 3k^2 - 4k - 3 = 0$

sum of all possible real values of k = 4/3

Q15. Let $A = \begin{bmatrix}\sin\theta & \cos\phi \\ -\cos\theta & \sin\phi\end{bmatrix}$ be an orthogonal matrix, where $\theta, \phi \in \mathbb{R}$. Then the maximum value of the expression $4\sin\theta + 3\cos\phi$ is :

(A) 1 (B) 7 (C) $\sqrt{7}$ (D) 5

Ans. D

Sol. Since A is an orthogonal matrix, we have $A^TA = I_2$.

$$A^TA = \begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\phi & \sin\phi \end{bmatrix} \begin{bmatrix} \sin\theta & \cos\phi \\ -\cos\theta & \sin\phi \end{bmatrix}$$

$$= \begin{bmatrix} \sin^2\theta + \cos^2\theta & \sin\theta\cos\phi - \cos\theta\sin\phi \\ \sin\theta\cos\phi - \cos\theta\sin\phi & \cos^2\phi + \sin^2\phi \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & \sin(\theta-\phi) \\ \sin(\theta-\phi) & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This implies $\sin(\theta-\phi) = 0$, so $\theta - \phi = n\pi$ for some integer n.

Thus, $\phi = \theta - n\pi$, which yields:

$\cos\phi = \cos(\theta - n\pi) = (-1)^n\cos\theta = \pm\cos\theta$

Now, substitute $\cos\phi$ into the given expression $E = 4\sin\theta + 3\cos\phi$:

$E = 4\sin\theta \pm 3\cos\theta$

The maximum value of $a\sin\theta + b\cos\theta$ is $\sqrt{a^2+b^2}$.

Hence, the maximum value of E is:

$E_{\max} = \sqrt{4^2 + (\pm 3)^2} = \sqrt{16+9} = 5$

Q16. If $x = \sin\theta + \cos\theta$ and $y = \sin 2\theta + \cos 2\theta$, where $\theta \in \mathbb{R}$, then the relation between x and y independent of θ is :

(A) $y^2 - 2y(x^2-1) + 2(x^2-1)^2 = 1$ (B) $y^2 - 2y(x^2+1) + 2(x^2+1)^2 = 1$

(C) $y^2 + 2y(x^2-1) + 2(x^2-1)^2 = 1$ (D) $y^2 - y(x^2-1) + (x^2-1)^2 = 1$

Ans. A

Sol. Given:

$x = \sin\theta + \cos\theta$

Squaring both sides:

$x^2 = 1 + 2\sin\theta\cos\theta = 1 + \sin 2\theta$

$\implies \sin 2\theta = x^2 - 1$

Now, using the second equation:

$y = \sin 2\theta + \cos 2\theta$

Substitute $\sin 2\theta = x^2 - 1$:

$y = (x^2-1) + \cos 2\theta$

$\implies \cos 2\theta = y - (x^2-1)$

Using the fundamental identity $\sin^2 2\theta + \cos^2 2\theta = 1$:

$(x^2-1)^2+(y-(x^2-1))^2=1$

Expanding the term:

$(x^2-1)^2+y^2-2y(x^2-1)+(x^2-1)^2=1$

$y^2-2y(x^2-1)+2(x^2-1)^2=1$

Q17. Consider the trigonometric equation $\sin(3x)=\frac{1}{2}$ for $x\in[0,2\pi)$. The total number of solutions of this equation that also satisfy the system of trigonometric inequalities $\sin x>\cos x$ and $\cos x<0$ is equal to :

(A) 1 (B) 3 (C) 2 (D) 4

Ans. C

Sol. First, we find all solutions of $\sin(3x)=\frac{1}{2}$ in the principal domain $x\in[0,2\pi)$.

Since $x\in[0,2\pi)$, we have $3x\in[0,6\pi)$.

$3x=\frac{\pi}{6},\frac{5\pi}{6},\frac{13\pi}{6},\frac{17\pi}{6},\frac{25\pi}{6},\frac{29\pi}{6}$

$x\in\left\{\frac{\pi}{18},\frac{5\pi}{18},\frac{13\pi}{18},\frac{17\pi}{18},\frac{25\pi}{18},\frac{29\pi}{18}\right\}$

$\because \cos x<0 \ and \ \sin x>\cos x$

$\Rightarrow x\in\left(\frac{\pi}{2},\frac{5\pi}{4}\right)\subset[0,2\pi)$

$\frac{\pi}{2}<\frac{13\pi}{18}<\frac{17\pi}{18}<\frac{5\pi}{4}$

$\Rightarrow x=\frac{13\pi}{18},\frac{17\pi}{18}$

Thus, there are exactly 2 solutions satisfying the given conditions.

Q18. Let A and B be two 3×3 real matrices such that A is a symmetric matrix and B is a skew-symmetric matrix. If $\mathrm{Tr}(A^2)=22$ and $\mathrm{Tr}((A+B)^2)=14$, then $\mathrm{Tr}(B^2)$ is equal to :

(A) 8 (B) -8 (C) 36 (D) -4

Ans. B

Sol. Given that $A^T=A$ and $B^T=-B$.

First, consider the trace of AB:

$\mathrm{Tr}(AB)=\mathrm{Tr}((AB)^T)=\mathrm{Tr}(B^TA^T)$

$=\mathrm{Tr}((-B)A)=-\mathrm{Tr}(BA)$

Since $\mathrm{Tr}(BA)=\mathrm{Tr}(AB)$, we have:

$\mathrm{Tr}(AB)=-\mathrm{Tr}(AB)\implies 2\mathrm{Tr}(AB)=0$

$\implies \mathrm{Tr}(AB)=0$

Now, expanding $(A+B)^2$:

$(A+B)^2 = A^2 + AB + BA + B^2$

Taking trace on both sides and using linearity of trace:

$\text{Tr}((A+B)^2) = \text{Tr}(A^2) + \text{Tr}(AB) + \text{Tr}(BA) + \text{Tr}(B^2)$

$\text{Tr}((A+B)^2) = \text{Tr}(A^2) + 2\text{Tr}(AB) + \text{Tr}(B^2)$

Substituting $\text{Tr}(AB) = 0$:

$\text{Tr}((A+B)^2) = \text{Tr}(A^2) + \text{Tr}(B^2)$

Given $\text{Tr}(A^2) = 22$ and $\text{Tr}((A+B)^2) = 14$:

$14 = 22 + \text{Tr}(B^2) \implies \text{Tr}(B^2) = -8$

Q19. The sum of all values of $x \in [0, 2\pi]$ satisfying the trigonometric equation $\sqrt{3}\sin 2x + \sqrt{3}\sin x = 1 + \cos 2x + \cos x$ and the system of inequalities $\tan x > 0$ and $\cos x < 0$ is equal to:

(A) 3π (B) $\frac{7\pi}{3}$ (C) $\frac{13\pi}{6}$ (D) $\frac{5\pi}{2}$

Ans. D

Sol. Given trigonometric equation:
$\sqrt{3}\sin 2x + \sqrt{3}\sin x = 1 + \cos 2x + \cos x$

Using the identities $\sin 2x = 2\sin x \cos x$ and $1 + \cos 2x = 2\cos^2 x$:
$\sqrt{3}\sin x(2\cos x + 1) = 2\cos^2 x + \cos x$
$\sqrt{3}\sin x(2\cos x + 1) = \cos x(2\cos x + 1)$
$(2\cos x + 1)(\sqrt{3}\sin x - \cos x) = 0$

This yields two cases for $x \in [0, 2\pi]$:
$2\cos x + 1 = 0 \implies \cos x = -\frac{1}{2}$

$\implies x = \frac{2\pi}{3}, \frac{4\pi}{3}$

$\sqrt{3}\sin x - \cos x = 0 \implies \tan x = \frac{1}{\sqrt{3}}$

$\implies x = \frac{\pi}{6}, \frac{7\pi}{6}$

$\Rightarrow x = \frac{4\pi}{3}, \frac{7\pi}{6}$

Sum of all valid solutions:
$\text{Sum} = \frac{7\pi}{6} + \frac{4\pi}{3} = \frac{15\pi}{6} = \frac{5\pi}{2}$

Q20. Let n be the number of digits in 6^{40} and m be the number of zeroes immediately after the decimal point before the first non-zero digit in $(0.125)^{20}$. Given $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, the value of $n - m$ is equal to:

(A) 14 (B) 13 (C) 15 (D) 12

Ans. A

Sol. To find n, the number of digits in 6^{40}:

$\log_{10}(6^{40}) = 40\log_{10}(2 \times 3)$

$= 40(\log_{10} 2 + \log_{10} 3)$

$\log_{10}(6^{40}) = 40(0.3010 + 0.4771)$

$= 40 \times 0.7781 = 31.124$

Characteristic of $\log_{10}(6^{40})$ is 31.
Thus, number of digits $n = 31 + 1 = 32$.
To find m, the number of zeroes in $(0.125)^{20}$:
$(0.125)^{20} = \left(\frac{1}{8}\right)^{20} = 2^{-60}$
$\log_{10}(2^{-60}) = -60\log_{10} 2$

$= -60 \times 0.3010 = -18.06 = -19 + 0.94$

Characteristic of $\log_{10}(0.125)^{20}$ is -19.

Thus, number of zeroes immediately after the decimal point before the first non-zero digit is $m = |-19| - 1 = 18$.

Hence, $n - m = 32 - 18 = 14$.

Q1. Let a, b, c be real numbers greater than 1 such that $\log_3 a$, $\log_3 b$, and $\log_3 c$ are in an Arithmetic Progression (AP) with a positive common difference. If a, b, c satisfy the relation $\log_a 3 + \log_c 3 = \frac{8}{3}\log_b 3$ and $a + c = 30$, then the value of $a + b + c$ is equal to _____.

Ans. 39

Sol. Let $\log_3 a = u - d$, $\log_3 b = u$, and $\log_3 c = u + d$, where $d > 0$ is the common difference of the AP.

Since $a, b, c > 1$, we have $u > d > 0$.

Using the change of base identity $\log_x y = \frac{1}{\log_y x}$, the given relation becomes:
$\frac{1}{\log_3 a} + \frac{1}{\log_3 c} = \frac{8}{3\log_3 b}$
Substitute the AP terms into the equation:
$\frac{1}{u-d} + \frac{1}{u+d} = \frac{8}{3u}$
$\frac{2u}{u^2 - d^2} = \frac{8}{3u}$
$6u^2 = 8(u^2 - d^2) \implies 2u^2 = 8d^2$
$\implies u^2 = 4d^2$
Since $u, d > 0$, taking the square root gives $u = 2d$.

Now, express $\log_3 a$, $\log_3 b$, and $\log_3 c$ in terms of d:

$\log_3 a = 2d - d = d \implies a = 3^d$

$\log_3 b = 2d \implies b = 3^{2d}$

$\log_3 c = 2d + d = 3d \implies c = 3^{3d}$

Given $a + c = 30$:

$3^d + 3^{3d} = 30$

Let $t = 3^d > 0$.

Then: $t^3 + t - 30 = 0$

$(t - 3)(t^2 + 3t + 10) = 0$

Since $t^2 + 3t + 10 = 0$ has a negative discriminant ($D = 9 - 40 = -31 < 0$), the only real root is $t = 3$.

Thus, $3^d = 3 \implies d = 1$.

Evaluating a, b, c:
$a = 3^1 = 3$
$b = 3^2 = 9$
$c = 3^3 = 27$
Therefore,
$a + b + c = 3 + 9 + 27 = 39$

Q2. For $r \in \mathbb{N}$, let D_r be the determinant defined by

$$D_r = \begin{vmatrix} 1 & r & r(r+1) \\ 1 & r+1 & (r+1)(r+2) \\ 1 & r+2 & (r+2)(r+3) \end{vmatrix}$$

If $S = \sum_{r=1}^{\infty} \frac{D_r}{r(r+1)(r+2)}$, then the value of $100S$ is equal to _____.

Ans. 50

Sol. Applying the row transformation $R_3 \to R_3 - 2R_2 + R_1$ to the determinant D_r:

$$D_r = \begin{vmatrix} 1 & r & r(r+1) \\ 1 & r+1 & (r+1)(r+2) \\ 0 & 0 & 2 \end{vmatrix} = 2$$

Now, the general term of the series is:

$$T_r = \frac{D_r}{r(r+1)(r+2)} = \frac{2}{r(r+1)(r+2)}$$

$$= \frac{(r+2) - r}{r(r+1)(r+2)} = \frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}$$

Summing the infinite telescopic series:

$$S = \sum_{r=1}^{\infty} T_r = \sum_{r=1}^{\infty} \left(\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \right)$$

$$= \frac{1}{1 \cdot 2} = \frac{1}{2}$$

Thus, $100S = 100 \times \dfrac{1}{2} = 50$.

Q3. For each integer $k \geq 1$, let $t_k = \tan\left(\dfrac{\theta_k}{2}\right)$, where $\theta_k \in \left(0, \dfrac{\pi}{2}\right)$. If $\dfrac{2t_k}{1+t_k^2} = \dfrac{4}{k(k+1)(k+2)}$ for all $k \geq 1$, then the value of $45 \times \displaystyle\sum_{k=1}^{8} \sin(\theta_k)$ is equal to _____.

Ans. 44

Sol. $\mathrm{T_k} = \sin\theta_k = \dfrac{2t_k}{1+t_k^2} = \dfrac{4}{k(k+1)(k+2)}$.

We split T_k into partial fractions using telescoping method:

$T_k = \dfrac{4}{k(k+1)(k+2)}$

$= 4 \cdot \dfrac{1}{2}\left[\dfrac{1}{k(k+1)} - \dfrac{1}{(k+1)(k+2)}\right]$

$= 2\left[\dfrac{1}{k(k+1)} - \dfrac{1}{(k+1)(k+2)}\right]$

Now, summing T_k from $k = 1$ to 8:

$S = \displaystyle\sum_{k=1}^{8} T_k = 2\sum_{k=1}^{8}\left[\dfrac{1}{k(k+1)} - \dfrac{1}{(k+1)(k+2)}\right]$

$S = 2\left[\left(\dfrac{1}{1\cdot 2} - \dfrac{1}{2\cdot 3}\right) + \left(\dfrac{1}{2\cdot 3} - \dfrac{1}{3\cdot 4}\right) + \cdots + \left(\dfrac{1}{8\cdot 9} - \dfrac{1}{9\cdot 10}\right)\right]$

$S = 2\left[\dfrac{1}{2} - \dfrac{1}{90}\right] = 2\left[\dfrac{44}{90}\right] = \dfrac{44}{45}$

Thus, the required value is:

$45 \times S = 45 \times \dfrac{44}{45} = 44$

Q4. If the sum of the infinite series $S = \displaystyle\sum_{n=1}^{\infty} \dfrac{3n+10}{n(n+1)(n+2)}$ is equal to L, then the value of L is _____.

Ans. 4

Sol. The general term of the series is:

$T_n = \dfrac{3n+10}{n(n+1)(n+2)}$

We express the numerator $3n+10$ in terms of factors of the denominator as $5(n+2) - 2n$.

Thus,

$T_n = \dfrac{5(n+2) - 2n}{n(n+1)(n+2)}$

$= \dfrac{5}{n(n+1)} - \dfrac{2}{(n+1)(n+2)}$

Splitting into telescopic differences :

$T_n = 5\left(\dfrac{1}{n} - \dfrac{1}{n+1}\right) - 2\left(\dfrac{1}{n+1} - \dfrac{1}{n+2}\right)$

Taking the sum from $n = 1$ to ∞:

$$S = 5\sum_{n=1}^{\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) - 2\sum_{n=1}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

$$S = 5(1) - 2\left(\frac{1}{2}\right) = 5 - 1 = 4$$

Hence, $L = 4$.

Q5. For a positive integer k, let T_k be defined by the determinant $T_k = \begin{vmatrix} k & k^2+1 & 2 \\ 1 & 1 & 2 \\ k+1 & k^2 & 2 \end{vmatrix}$ If $S_n = \sum_{k=1}^{n} T_k$, then the value of S_{10} is equal to _____.

Ans. 860

Sol. The determinant T_k can be evaluated as follows:

$R_2 \to R_2 - R_1$

$R_3 \to R_3 - R_1$

$T_k = 2k^2 + 2k - 2$

Now, calculating S_{10}:

$$S_{10} = \sum_{k=1}^{10} T_k = \sum_{k=1}^{10}(2k^2 + 2k - 2)$$

$$S_{10} = 2\sum_{k=1}^{10} k^2 + 2\sum_{k=1}^{10} k - \sum_{k=1}^{10} 2$$

= 770 + 110 – 20 = 860.